

# Kalen Cole Jaroszewski

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## EDUCATION

Texas A&M University – College Station, Texas      BS in Mechanical Engineering      GPA - 3.57      May 2027

## EXPERIENCES

**Human-Empowering Robotics and Control (HERC) Lab**      September 2025 – Present  
Undergraduate Researcher | Part-time

- Applying beam theory in the design of a 2-DOF flexural hinge with a 4 mm travel diameter, while withstanding a 230N radial force in a magnetic cycloidal gearbox (CyMG)
- Performing an adhesive performance trade study to address bonding failures to nickel-coated magnets
- Conducted FEA to develop a dynamometer rated up to 150 N m, ensuring compatibility with the lab's actuators

**Texas A&M Engineering Experiment Station - RAD Lab**      June 2025 – August 2025  
Mechanical Intern

- Presented a trade study focused on improving torque density, while reducing stiction and hysteresis in CyMGs
- Manufactured a 20:1 Halbach-array-based CyMG compatible with both traditional C-bearing and experimental blade flexure rotor design to enable stiction analysis
- Designed a RESOLUTE encoder readhead mount capable of adjusting within its 0.7–0.9 mm operating range using finite element analysis and hand calculation validation

## ORGANIZATIONS

**TAMU AggieSat Laboratory** - AggieSat8 TMS Member | Team of 6      November 2025 – Present

- Creating trade studies for the thermal, mechanisms, and structures of an ISS experiment payload, validating the space-worthiness of liquid crystal antennas
- Applying the V-model systems engineering life cycle, from requirements and verification to validation

**Texas A&M University Robotics Team and Leadership Experience**      January 2024 – Present  
Development Vice President | Senior leadership, 400 members      May 2024 – Present

- Engaging 200+ undergraduates a week by coordinating the Hatchling Development Program and managing the 20 inter-organizational officers through a data-driven sustainability framework
- Doubled Hatchling reach and internal capacity limitations by establishing and executing a multi-organization expansion plan, resulting in partnerships with 3 TAMU and TAMU Galveston organizations
- Achieved a 2,373% increase in applications (to 356), a sixfold rise in project completion (to 60%), and net-positive operations within three semesters by spearheading a comprehensive overhaul of Hatchling
- Implemented an automated Excel tracking system, reducing procurement lead time by 1 week across 21 project teams, and enabling spending analysis to drive budget allocation discussions

**QUAD Mechanical Lead** | Team of 10      August 2024 – December 2025

- Led the mechanical design of two Raspberry Pi quadruped robots, including QUADV2 (\$2,400, 3-foot-long dog) and Mini-Quad (sub \$100, 10-inch-long dog) serving as a software testbed for QUADV2
- Engineered a backdrivable 33 N m cycloidal actuator meeting safety and torque feedback control requirements using an MJ5208 DC motor controlled via Moteus-C1/R4 boards
- Designed a 3-DOF robotic leg with 115.2° knee ROM, optimized for additive manufacturing and assembly, featuring modular actuators, replaceable sensor-embedded silicone feet, and a toe-link failure mode

**TAMU IGNITORS Rocketry Team** - Avionics Member | Team of 15      November 2024 – May 2025

- Manufactured all SRAD flight-critical components via a CO2 Laser Cutter and additive manufacturing

## SKILLS

- **Mechanical:** SolidWorks (CSWP), ANSYS FEA, GD&T Y14.5, DFMA, Soldering, Power Tools
- **Software:** MS Office Suite (Excel, Word, PowerPoint), Python, Git, GitHub, Linux, Inkscape